Network Reconnaissance & Footprinting

Minor Project – Cyber Security Internship

Field	Detail
Prepared by	Alen Shibu
Target	scanme.nmap.org (authorized public test host operated by the Nmap Project)
Environment	Kali Linux (attacker workstation) in Oracle VirtualBox VM
Project folder	~/InternVision-Mini-Project
Date of testing	October 8, 2026
Project type	Minor Project – Network Reconnaissance & Footprinting (InternVision Cyber Security Internship)

1. Project Goal

Gather maximum publicly available information about a pre-approved target domain and map its public-facing network presence, using primarily passive, publicly available information sources, supplemented by limited and explicitly authorized active probing against a host provided for that purpose.

2. Scope & Legal Note

Passive, non-intrusive lookups (WHOIS, DNS enumeration, Google dorking) were performed against the parent domain nmap.org — the Nmap Project's own live public website — since this information is intentionally public and no requests beyond ordinary lookups were made to it. All active probing (ping, traceroute, Nmap port/service scanning, curl, and WhatWeb) was limited strictly to scanme.nmap.org, a host the Nmap Project explicitly provides and authorizes for this kind of testing. No destructive, exploitative, or denial-of-service testing was performed against any host. scanme.nmap.org is explicitly designated by the Nmap Project for scan testing, satisfying the project brief's requirement of a target "explicitly designated for testing."

3. Tools Used

Tool	Purpose
whois	Domain registration lookup
dig / nslookup	DNS record enumeration (A, AAAA, MX)
Google Search (dorking)	Discovery of publicly indexed files/pages
ping	Reachability and latency testing
traceroute	Network path mapping
nmap	Port scanning and service/version detection
curl	HTTP header banner grabbing
whatweb	Web technology fingerprinting

4. Phase 1 – Public Information Gathering

4.1 WHOIS Lookup

Since scanme.nmap.org is a subdomain, the WHOIS lookup was performed against the registered parent domain, nmap.org, directly via the .org registry's WHOIS server (whois.pir.org).

```
whois -h whois.pir.org nmap.org
```

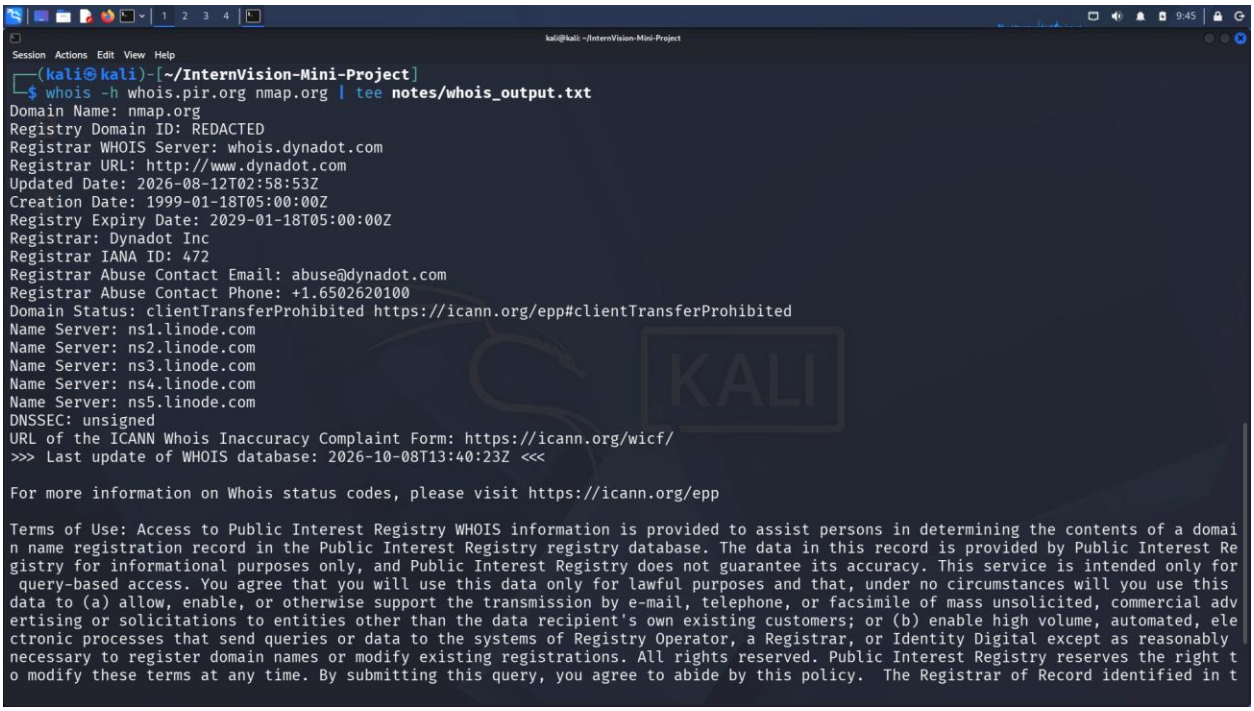


Figure 1: WHOIS registration details for nmap.org

Field	Value
Domain Name	nmap.org
Registrar	Dynadot Inc
Creation Date	1999-01-18
Expiry Date	2029-01-18
Name Servers	ns1–ns5.linode.com
DNSSEC	Unsigned

Note: the WHOIS output includes no registrant, administrative, or technical contact details; the only contact information shown is the registrar's abuse-report address. The Registry Domain ID field is marked REDACTED. This is normal for how registrars publish WHOIS data and is not itself a finding.

Finding: The domain is long-established (registered in 1999) and its DNS is hosted on Linode's name servers. DNSSEC is not enabled, meaning DNS responses for this domain are not cryptographically signed. This is an observation about the domain's DNS configuration, not a confirmed vulnerability — the large majority of domains on the internet operate without DNSSEC.

4.2 DNS Enumeration

DNS records were queried using both dig and nslookup, pointed explicitly at Google's public resolver (8.8.8.8) after the VM's default local resolver (192.168.1.1) failed to respond.

```
dig @8.8.8.8 scanme.nmap.org A +short
dig @8.8.8.8 nmap.org MX
dig @8.8.8.8 scanme.nmap.org ANY
nslookup scanme.nmap.org 8.8.8.8
```

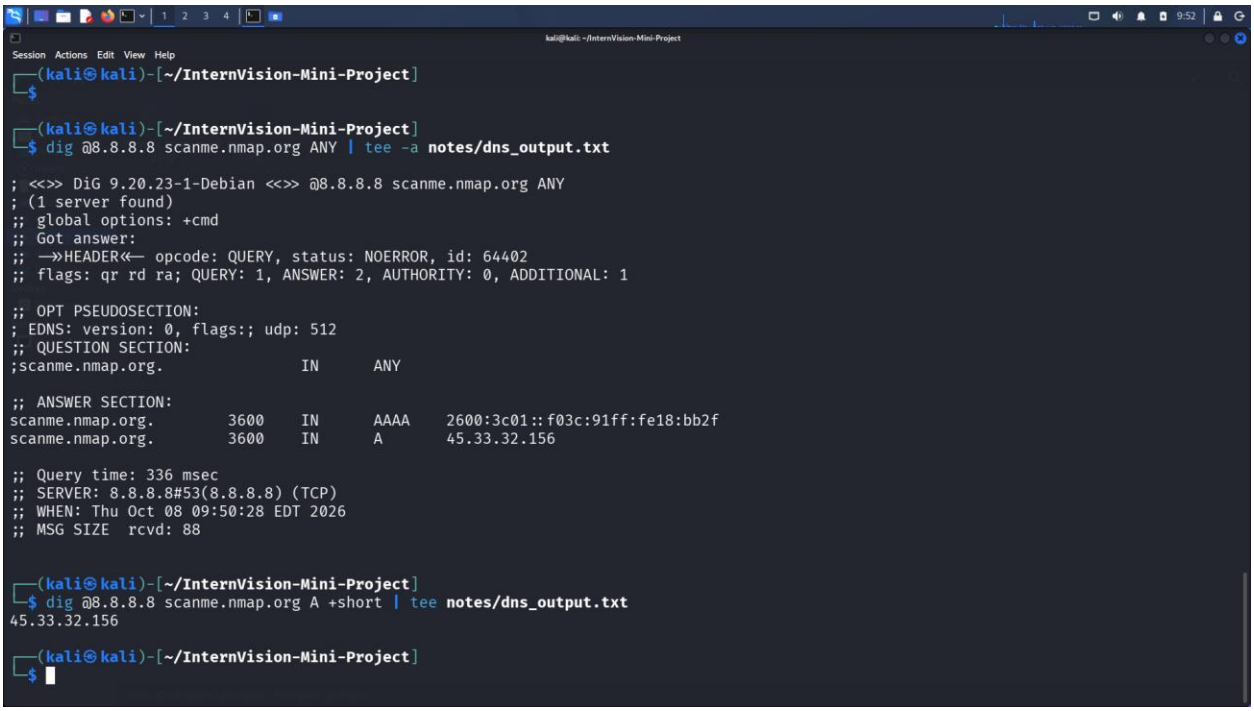


Figure 2: A record lookup for scanme.nmap.org

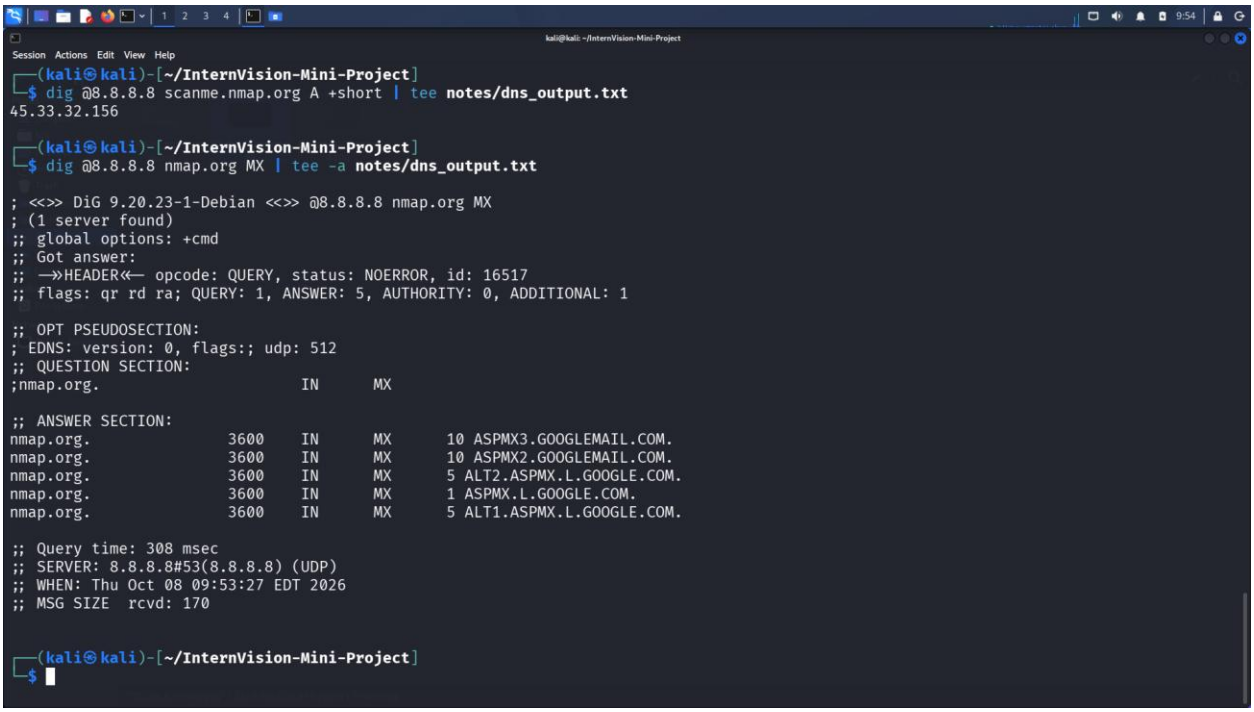


Figure 3: MX record lookup for nmap.org

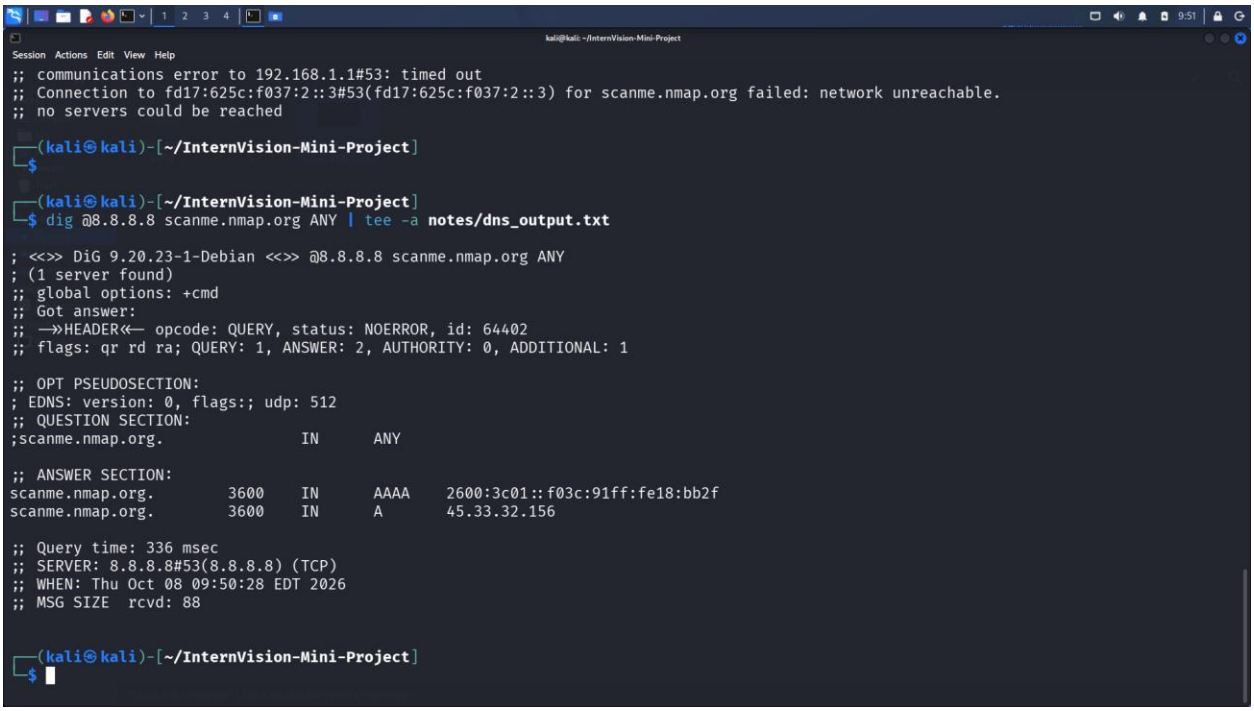


Figure 4: ANY query result (A and AAAA only) for scanme.nmap.org

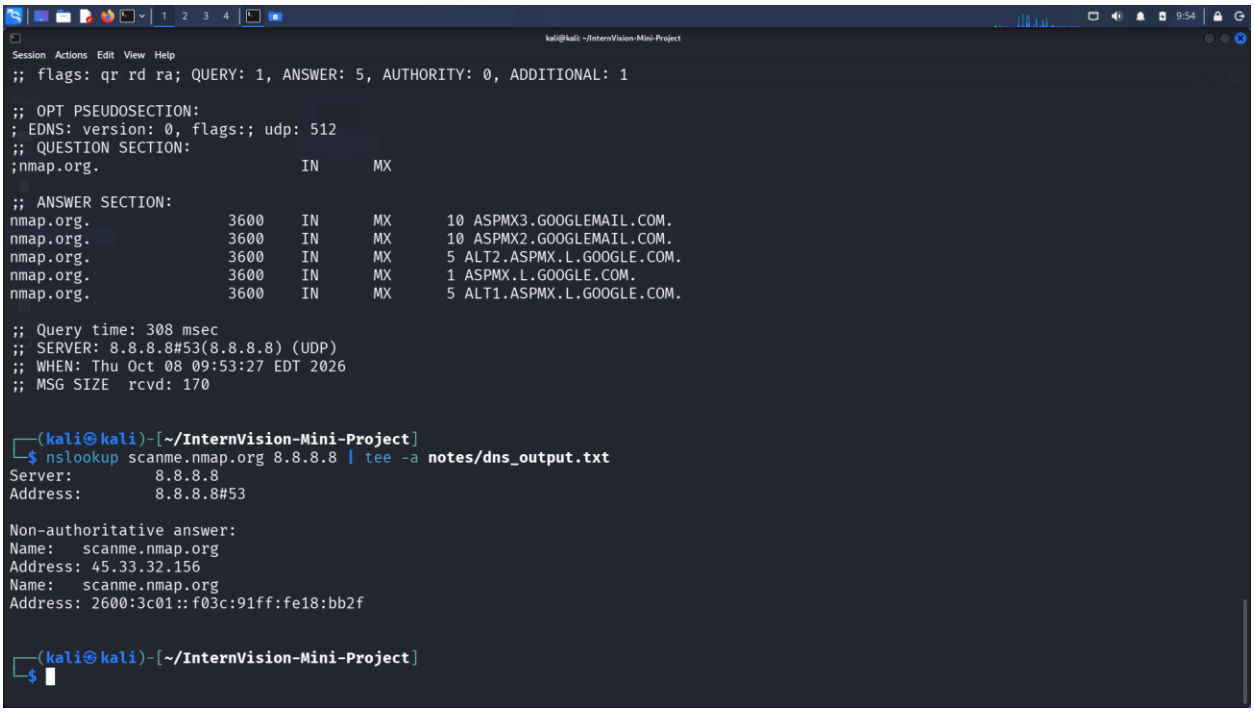


Figure 5: nslookup cross-check against Google DNS

Record Type	Value
A (IPv4)	45.33.32.156
AAAA (IPv6)	2600:3c01::f03c:91ff:fe18:bb2f
MX	ASPMX.L.GOOGLE.COM (priority 1), ALT1/ALT2.ASPMX.L.GOOGLE.COM (priority 5), ASPMX2/ASPMX3.GOOGLEMAIL.COM (priority 10)

Finding: The MX records indicate that incoming email for nmap.org is routed through Google's mail servers. The results also show that DNS nameserver hosting and mail routing use different providers (nameservers on Linode, mail on Google). (Note: nameserver hosting and web-server hosting are separate things; this lookup does not establish who hosts the actual website content.)

Note: a dig ANY query against a public resolver typically returns only a minimal answer — here, just the A and AAAA records (Figure 4) — rather than every record type the domain has. NS, TXT, and SOA records were not separately queried in this pass and would need dedicated lookups for a more complete picture.

4.3 Subdomain Enumeration

```
for sub in www mail ftp admin dev test; do echo "== $sub =="; dig @8.8.8.8 $sub.nmap.org
+short; done
```

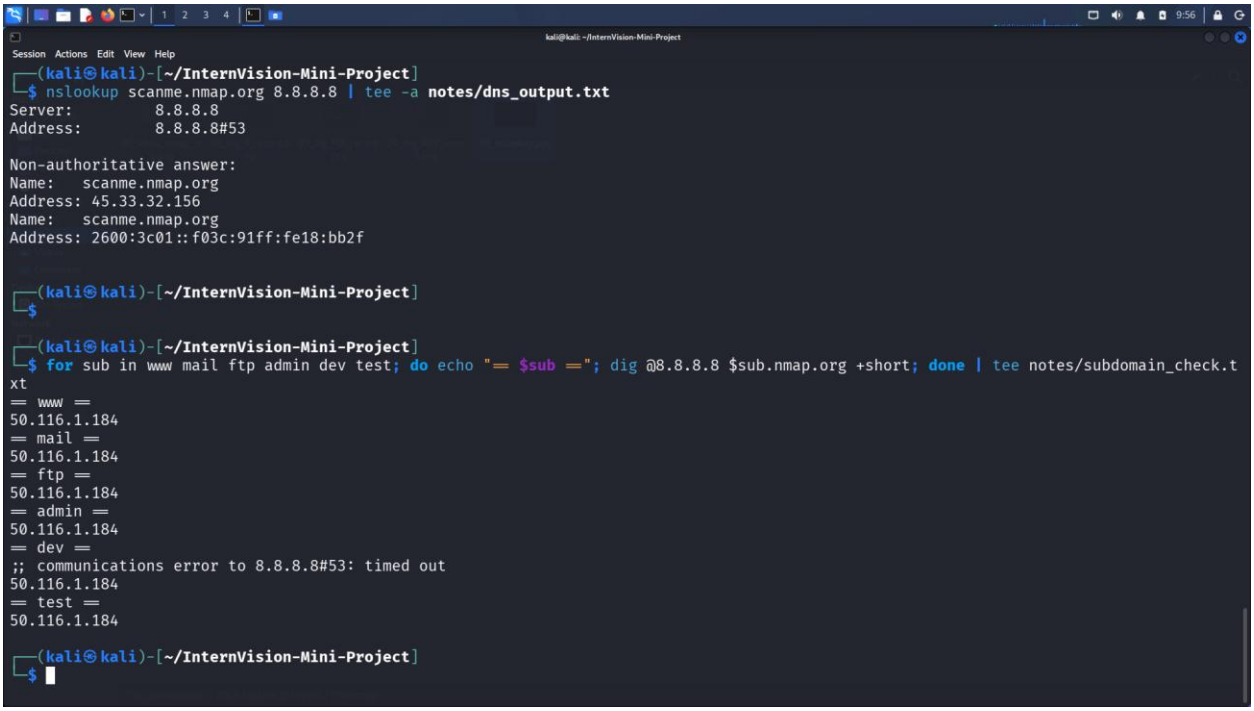


Figure 6: Subdomain resolution check for common hostnames under nmap.org

Finding: Every tested subdomain (www, mail, ftp, admin, dev, test) resolved to the identical IP address 50.116.1.184, which is suggestive of a wildcard DNS record. However, this was not formally confirmed: a proper wildcard test requires querying a hostname guaranteed not to exist and checking whether it also resolves, and that control test was not performed in this pass. Until that check is done, the wildcard explanation should be treated as a plausible but unconfirmed hypothesis rather than a settled finding. It's also worth noting that the dev lookup (Figure 6) initially returned a timeout before succeeding on an automatic retry — a minor network blip, not a DNS result in itself.

Deliverable note: of the hostnames checked, only scanme.nmap.org itself is a confirmed, individually-verified live subdomain (it is the actual target used throughout Phase 2). The remaining names (www, mail, ftp, admin, dev, test)

cannot be individually confirmed as genuinely existing services, precisely because of the suspected wildcard behavior above.

4.4 Google Dorking

The following dork queries were run manually in a browser against nmap.org to check for unintentionally exposed files or pages:

- site:nmap.org filetype:pdf
- site:nmap.org filetype:xlsx
- site:nmap.org inurl:login
- site:nmap.org intitle:"index of"

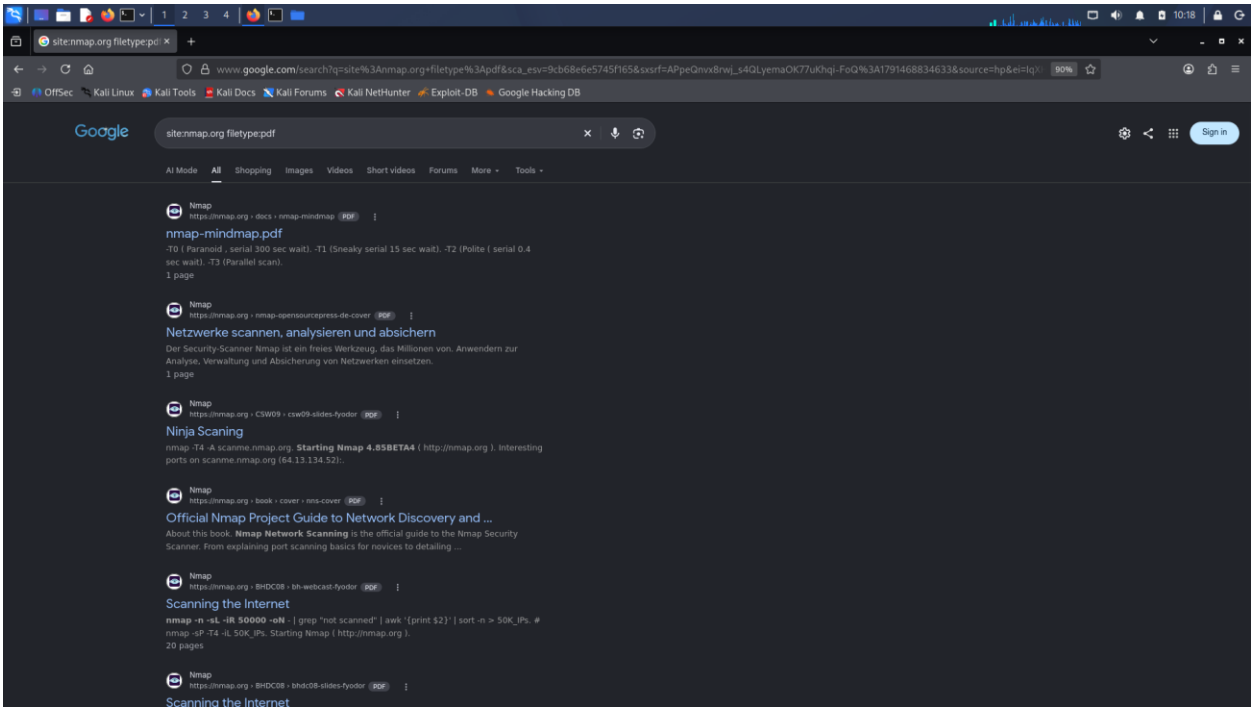


Figure 7: site:nmap.org filetype:pdf — multiple indexed PDFs (documentation, slide decks)

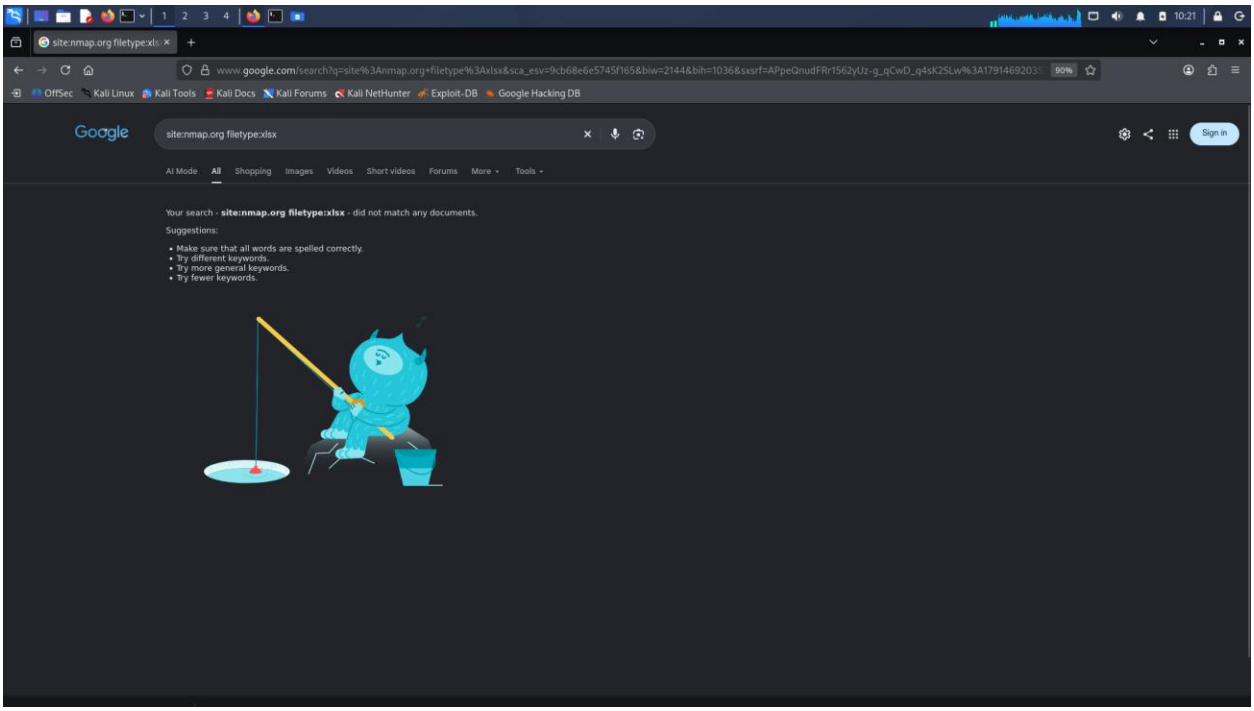


Figure 8: site:nmap.org filetype:xlsx — no results found

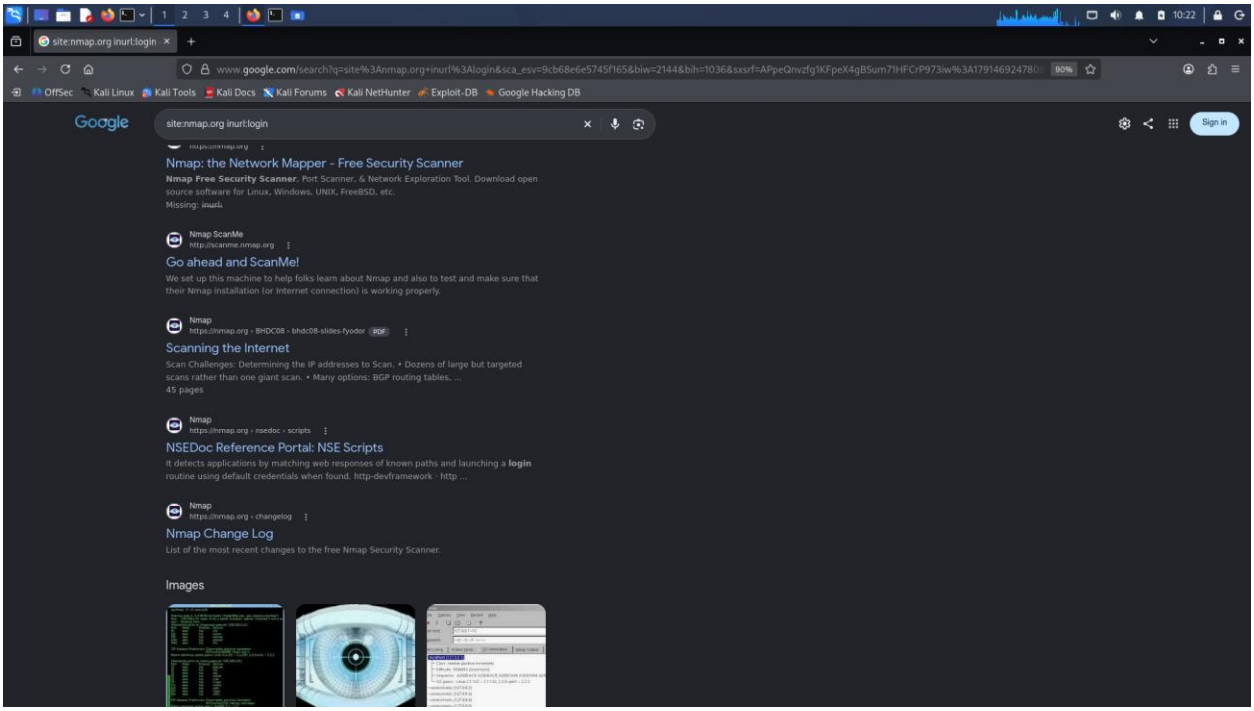


Figure 9: site:nmap.org inurl:login — no dedicated login pages indexed

Caveat: Google's results page shows "Missing: inurl:" beneath the first result, meaning Google did not actually apply the inurl: operator to this query and fell back to a plain keyword match instead. This result is therefore indicative only, not confirmation that no login-related URLs exist on the site.

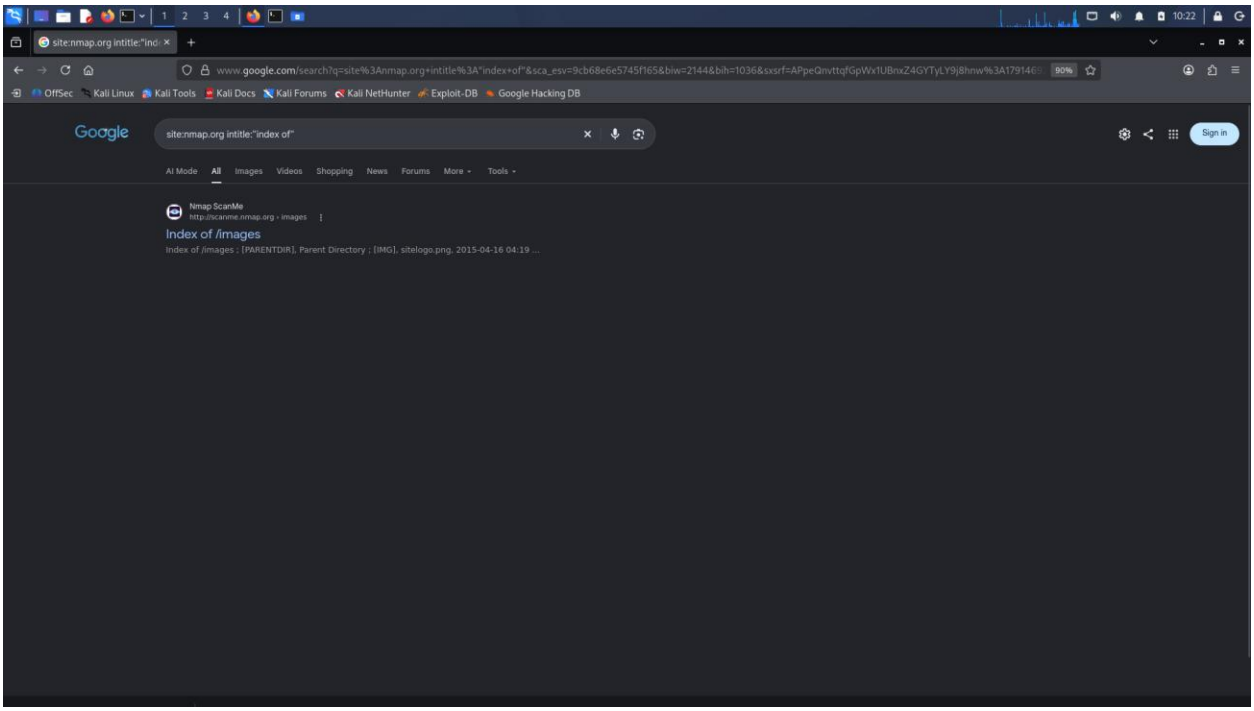


Figure 10: site:nmap.org intitle:"index of" — one directory-listing page indexed by Google (scanme.nmap.org/images)

Dork	Result
filetype:pdf	Multiple PDFs indexed; displayed results appeared to be public documentation or slides. File contents were not exhaustively reviewed.
filetype:xlsx	No results returned for the tested query at the time of testing.
inurl:login	The search did not conclusively establish whether login-related URLs exist; Google did not apply the intended inurl: operator as expected.
intitle:"index of"	Google has indexed a directory-listing page for scanme.nmap.org/images

Finding: No sensitive information was apparent in the displayed search results; this was not a comprehensive content review. Google has indexed a directory-listing page at scanme.nmap.org/images; this confirms the page was indexed with that title, but does not by itself prove the listing is still live or that its contents were independently reviewed — the URL was not browsed directly as part of this exercise. If confirmed live, an open directory listing exposing only non-sensitive image assets would be a low-severity informational finding (disabling directory listing is a standard hardening measure regardless).

5. Phase 2 – OS / Network Fingerprinting

5.1 Ping Test

```
ping -c 4 scanme.nmap.org
```

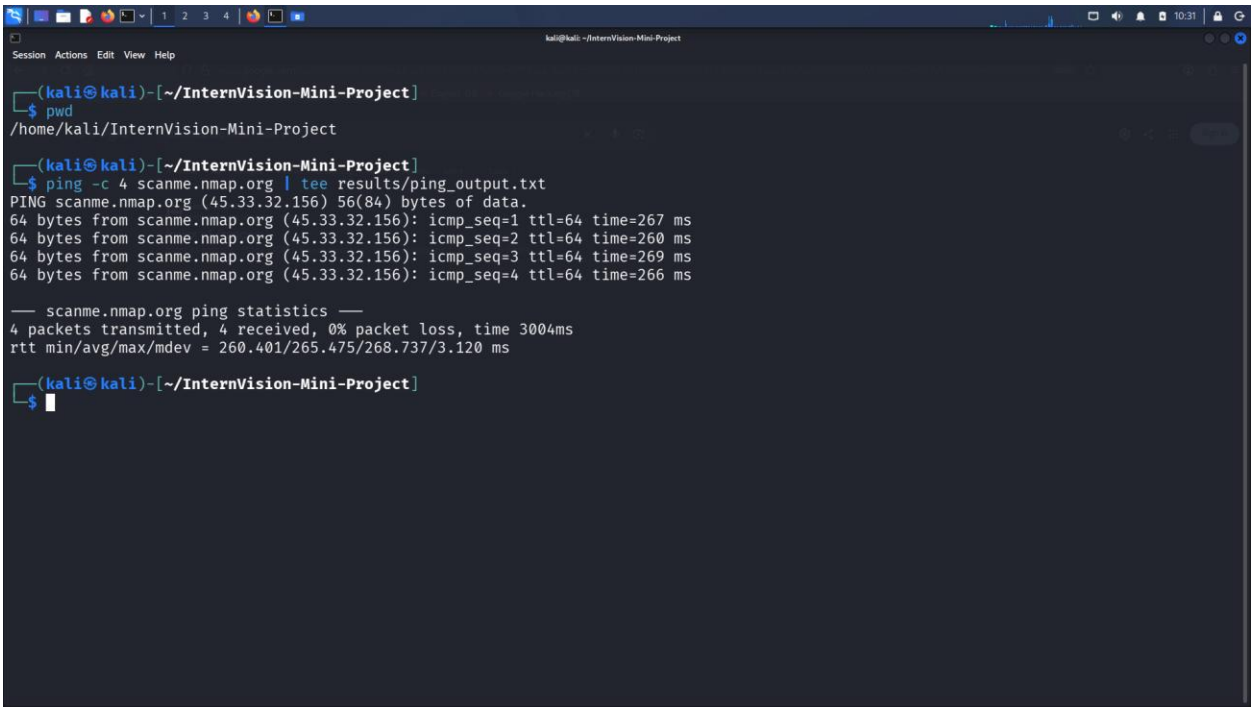


Figure 11: ICMP ping test to scanme.nmap.org

Result: Four replies were received, with 0% packet loss and an average round-trip time of approximately 265 ms. This confirms that the host responded to ICMP Echo Requests during the test. The measured latency alone does not establish the server's geographic location.

5.2 Traceroute

```
traceroute scanme.nmap.org
```

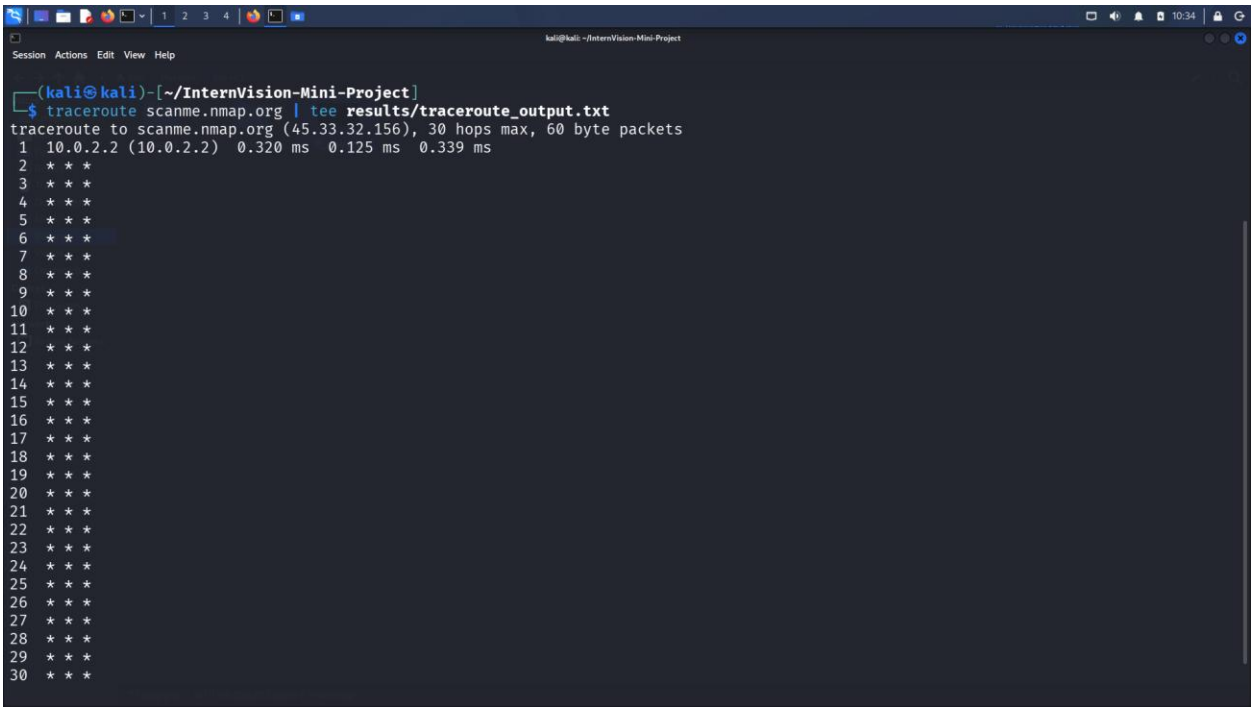


Figure 12: Traceroute path to scanme.nmap.org

Result: Hop 1 (10.0.2.2) is VirtualBox's NAT gateway for the VM. All subsequent hops (2–30) returned no response (* *). Because this VM is running in NAT mode, the most likely explanation is that the NAT layer itself does not relay the ICMP time-exceeded replies back to the guest — rather than every router beyond the gateway independently choosing to filter probes. Filtering by intermediate internet routers remains a secondary possibility and cannot be ruled out from this evidence alone. Either way, the destination was confirmed reachable via the ping test above. The ping replies consistently displayed a TTL value of 64; this is recorded as an observation only and does not establish the destination's geographic location or network distance.*

5.3 Service & Version Detection (Nmap)

```
sudo nmap -sV -T4 --top-ports 100 scanme.nmap.org
```

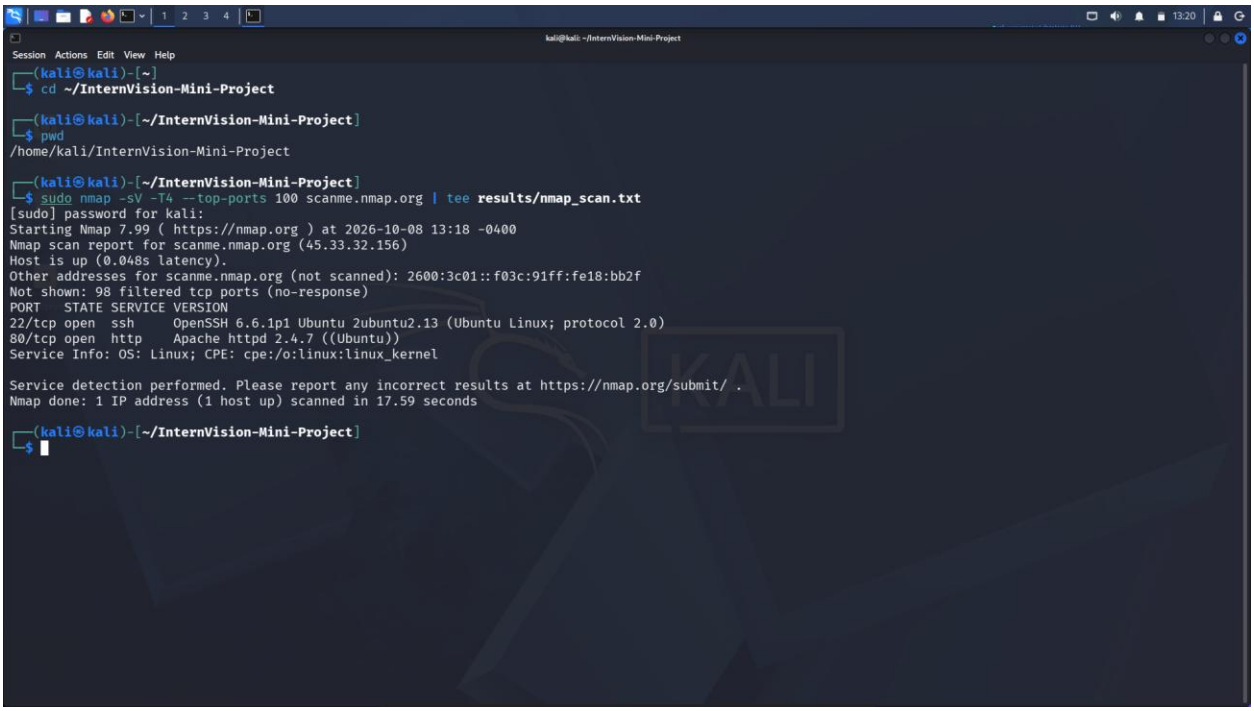


Figure 13: Nmap service/version detection scan results

Port	State	Service	Version
22/tcp	open	ssh	OpenSSH 6.6.1p1 (Ubuntu 2ubuntu2.13)
80/tcp	open	http	Apache httpd 2.4.7 (Ubuntu)
443/tcp	included in “98 filtered”	https	see discrepancy note below

Note on port 443: the scan's “Not shown: 98 filtered tcp ports (no-response)” line means port 443 (one of the top-100 ports scanned) falls into that filtered bucket by elimination, but it was not individually itemized in this output. Separately, the WhatWeb probe to port 443 (Figure 15, Section 5.5) returned “Connection refused” rather than a timeout — a refused connection normally indicates an actively closed port (a RST response), which is a different signal from a filtered, silently-dropped one. This is a minor inconsistency between the two tools' observations that could not be fully reconciled with the evidence collected in this pass; a dedicated single-port check (e.g. nmap -Pn -p 443 scanme.nmap.org) would resolve it in a follow-up test.

Operating System (inferred): Linux — specifically Ubuntu. Nmap's Service Info line reports only the general OS family (“OS: Linux”) from service-detection fingerprinting; the more specific “Ubuntu” identification comes directly from the OpenSSH and Apache version banners themselves (both explicitly state “Ubuntu”), which curl and WhatWeb also read from the same server responses. This is a banner-based inference, not a dedicated OS fingerprint — Nmap's dedicated OS-detection scan (nmap -O) was not run as part of this exercise.

Finding: The detected OpenSSH and Apache version strings are notably old. This is worth noting as it narrows the search space for known CVEs an attacker might try, but it is not by itself proof the host is actually vulnerable or unpatched — many Linux distributions, including Ubuntu, backport security fixes into existing packages without changing the reported version string, so a banner alone cannot confirm or rule out patch status.

Note: Nmap reported a host latency of approximately 48 ms in Figure 13, whereas the separate ping test in Section 5.1 reported an average RTT of approximately 265 ms. These measurements were obtained using different probes and at different moments. The available evidence does not establish the cause of the difference; both tests nevertheless recorded the host as reachable.

5.4 HTTP Header Banner Grab

```
curl -I http://scanme.nmap.org
```

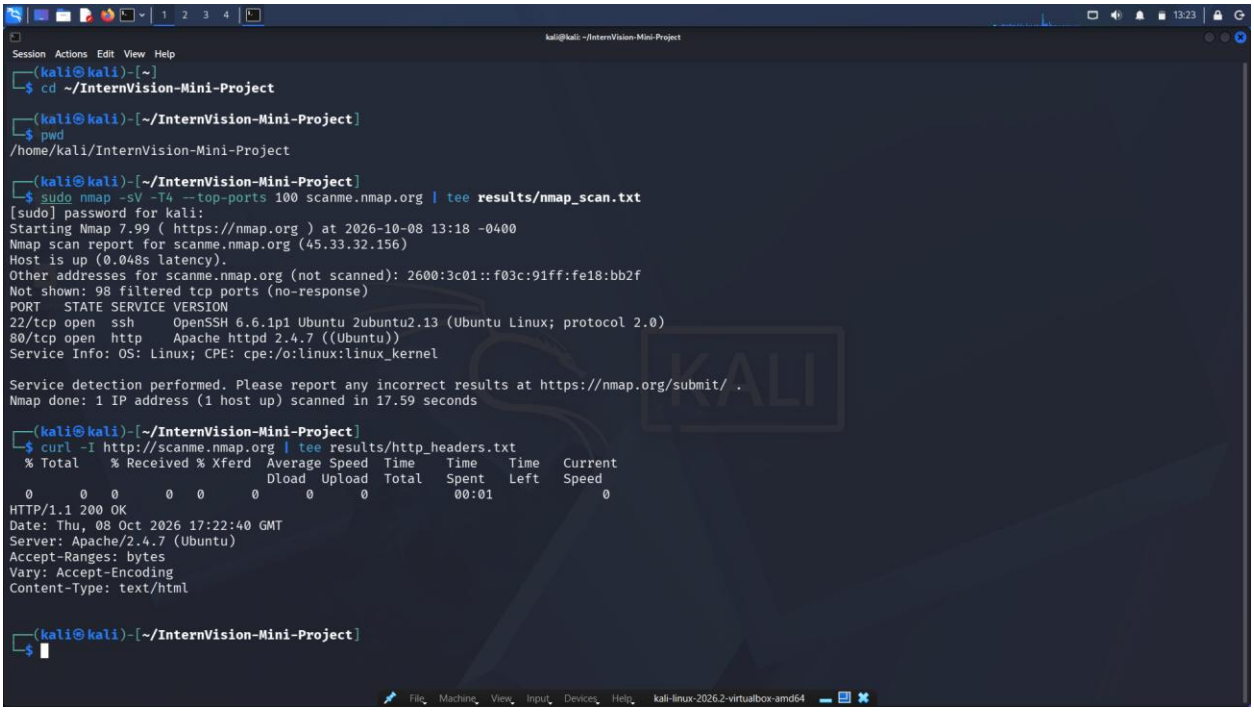


Figure 14: HTTP response headers from the web server

Result: Server header confirms Apache/2.4.7 (Ubuntu), matching the Nmap version-detection result exactly.

5.5 WhatWeb Fingerprint

```
whatweb scanme.nmap.org
```

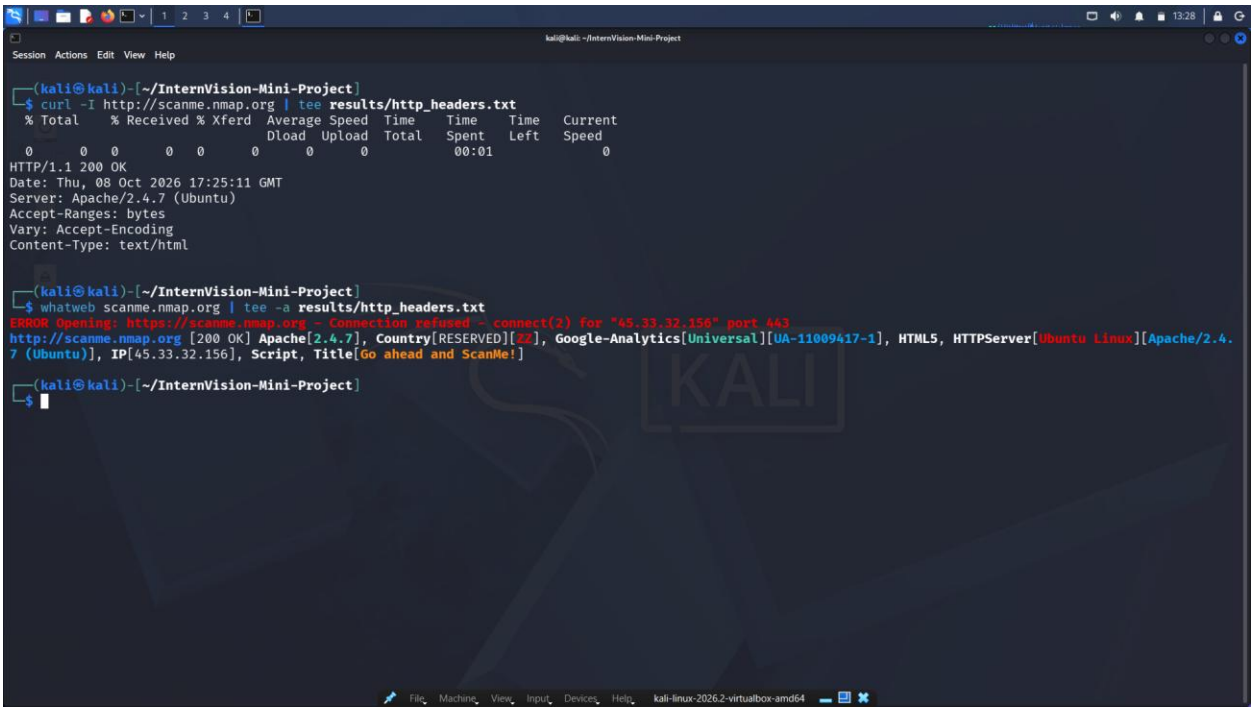


Figure 15: WhatWeb technology fingerprint

Result: WhatWeb's HTTPS probe failed with “Connection refused” on port 443, which normally indicates an actively closed port rather than a filtered one — this differs from Nmap's result and is discussed as an unreconciled discrepancy in Section 5.3. WhatWeb automatically fell back to HTTP (port 80), confirming: Apache 2.4.7, Ubuntu Linux, HTML5, and a Google Analytics tracking tag (UA-11009417-1) embedded in the page.

Note: as shown in Figure 15, the WhatWeb command used tee -a (append) rather than plain tee, so this output was added to the existing results/http_headers.txt file — which already held the curl output from Section 5.4 — instead of overwriting it. Both outputs are preserved in that file.

6. Summary of Findings

#	Data Point	Tool(s) Used	Result
1	Domain registration	whois	nmap.org — registered 1999, Dynadot, NS on Linode
2	IP addresses	dig, nslookup	IPv4: 45.33.32.156 / IPv6: 2600:3c01::f03c:91ff:fe18:bb2f
3	Mail servers	dig (MX)	Google Workspace (ASPMX.L.GOOGLE.COM and alternates)
4	Subdomains	dig (loop)	All test names resolve to 50.116.1.184 — suggestive of wildcard DNS, not formally confirmed; only scanme.nmap.org individually verified
5	Indexed files/pages	Google dorking	Multiple PDFs indexed (contents not exhaustively reviewed); one indexed image-directory listing (not independently browsed); no spreadsheets returned; login-URL query inconclusive (operator not applied by Google)
6	Reachability / latency	ping	Reachable, 0% loss, ~265 ms average RTT
7	Network path	traceroute	Hops unresponsive past the VM's NAT gateway — most likely NAT behavior, router filtering not ruled out; destination still confirmed reachable
8	Open ports / services	nmap	22/tcp SSH (OpenSSH 6.6.1p1), 80/tcp HTTP (Apache 2.4.7); 443/tcp within the 98 filtered ports (not individually listed) per Nmap, but "connection refused" per WhatWeb — unreconciled discrepancy
9	Operating system	nmap, curl, whatweb	Linux — Ubuntu, inferred from service banners (not a dedicated -O OS fingerprint)

7. Conclusion

Using primarily passive, publicly available information sources — supplemented by limited, explicitly authorized active probing against scanme.nmap.org — this exercise mapped most of the public-facing footprint of the target: registration details, DNS infrastructure, mail provider, open services, and a banner-based operating system inference were all identified without sending any exploit or attack traffic. A few results remain open rather than fully settled and are flagged rather than overstated in this report: the suspected wildcard DNS behavior under nmap.org was not confirmed with a proper control test, the unresponsive traceroute hops are most likely explained by the VM's NAT networking rather than confirmed router filtering, and Nmap's and WhatWeb's observations on port 443 disagree and were not reconciled. None of these open items change the overall picture, but they would be the natural next steps in a follow-up test.